SUPPLEMENTARY DATA 1: SEARCH STRATEGY

Databases: MEDLINE (via Ovid), EMBASE, Web of Science and IEEEXplore

MEDLINE = 429

“Deep learning” [MESH] or "Machine Learning"[Mesh] OR "Artificial Intelligence"[Mesh] OR "Neural Networks (Computer)"[Mesh] OR "Support Vector Machine"[Mesh] OR “Algorithms” [MESH] OR “Bayes theorem” [MESH] OR Markov Chains[MESH] OR Machine learning OR Artificial Intelligence OR Naive Bayes OR Neural network* OR support vector machine OR SVM OR random forest* OR boosting OR adaboost OR deep learning OR (deep or convolutional or bayesian) network* OR bayes classifier OR machine intelligence OR computational intelligence OR or (nearest adj1 (neighbor or neighbour)) or “multilayer perceptron” OR computer vision OR hidden markov mode* OR hmm

AND

Surgical performance OR surgical skill OR surgical assessment OR technical skill OR technical performance OR technical assessment OR surgeon* adj3 performance OR surgeon* adj3 assess* OR surgeon* adj3 skill

Embase = 388

(“Deep learning” [mesh]" OR "Machine Learning"[mesh]" OR "Artificial Intelligence" [mesh]" OR "artificial Neural Network[mesh]" OR convolutional neural network [mesh]"

"Support Vector [mesh]" " OR Algorithm [mesh] OR " Bayes theorem" [mesh] OR markov chain[mesh] or OR Machine learning OR Artificial Intelligence OR Naive Bayes OR Neural network* OR support vector machine OR SVM OR random forest* OR boosting OR adaboost OR deep learning OR (deep or convolutional or neural or bayesian) network* OR bayes classifier OR machine intelligence OR computational intelligence OR or (nearest adj1 (neighbor or neighbour)) or “multilayer perceptron” OR computer vision OR hidden markov mode* OR hmm

AND

Surgical performance OR surgical skill OR surgical assessment OR technical skill OR technical performance OR technical assessment OR surgeon* NEAR/3 performance OR surgeon* NEAR/3 assess* OR surgeon* NEAR/3 skill

Web of Science = 595

TS = (Skill Assess* OR Performance assess* OR technical assess*) 445,970

AND

TS = (surgeon OR surgery OR surgical) 1,568,312

AND

TS = (Deep Learning OR Machine learning Or Artificial intelligence OR neural network* OR computer vision OR support vector machine OR Bayesian OR markov chain* OR hidden markov model* OR adaboost) 544,866

1,2,3 = 595

IEEEExplore = 484

(("Abstract": performance OR "Abstract": skill OR "Abstract": assess) AND ("Abstract": surgeon OR "Abstract": surgery OR "Abstract": surgical) AND (machine learning OR automat OR artificial intelligence OR computer vision))

SUPPLEMENTARY DATA 2: MERSQI QUALITY ASSESSMENT

Author	Year	Study Design /3	Sampling /3	Type of Data /3	Validity of evaluation instrument /3	Data analysis /3	Outcomes /3	Total/ 18
BENCHTOP								
Rosen et al[76]	2001	1	2	3	0	3	1.5	**10.5**
Rosen et al[77]	2001	1	2	3	0	3	1.5	**10.5**
Leong et al[78]	2007	1	2	3	2	3	1.5	**12.5**
Varadarajan et al[30]	2009	1	2	3	0	3	1.5	**10.5**
Reiley et al[31]	2009	1	2	3	0	3	1.5	**10.5**
King et al[18]	2009	1	2	3	0	3	1.5	**10.5**
Islam et al[58]	2011	1	2	3	0	3	1.5	**10.5**
Tao et al[32]	2012	1	2	3	0	3	1.5	**10.5**
Kumar et al[33]	2012	1	3	3	3	3	1.5	**14.5**
Ahmidi et al[26]	2012	1	2	3	0	3	1.5	**10.5**
Islam et al[59]	2013	1	2	3	0	3	1.5	**10.5**
Ahmidi et al[34]	2013	1	2	3	0	3	1.5	**10.5**
Oropesa et al[19]	2013	1	2	3	0	3	1.5	**10.5**
Sharma et al[64]	2014	1	2	3	2	3	1.5	**12.5**
Weede et al[20]	2014	1	2	3	1	3	1.5	**11.5**
Watson[27]	2014	1	2	3	1	3	1.5	**11.5**
Zia et al[65]	2015	1	2	3	3	3	1.5	**13.5**
Islam et al[60]	2016	1	2	3	3	3	1.5	**13.5**
Yamaguchi et al[61]	2016	1	2	3	0	3	1.5	**10.5**
Zia et al[66]	2016	1	2	3	2	3	1.5	**12.5**
Dockter et al[22]	2017	1	2	3	0	3	1.5	**10.5**
Forestier et al[35]	2017	1	2	3	2	3	1.5	**12.5**
Jiang et al[46]	2017	1	2	3	0	3	1.5	**10.5**
French et al[21]	2017	1	2	3	2	3	1.5	**12.5**
Brown et al[36]	2017	1	2	3	3	3	1.5	**13.5**
Sun et al[28]	2017	1	2	3	0	3	1.5	**10.5**
Zia et al[80]	2018	1	2	3	3	3	1.5	**13.5**

Zia et al[37]	2018	1	2	3	2	3	1.5	**12.5**
Miller et al[67]	2018	1	2	3	0	3	1.5	**10.5**
Uemura et al[23]	2018	1	2	3	0	3	1.5	**10.5**
Sgouros et al[62]	2018	1	2	3	0	3	1.5	**10.5**
Oquendo et al[24]	2018	1	2	3	3	3	1.5	**13.5**
Wang et al[38]	2018	1	2	3	2	3	1.5	**12.5**
Wang et al[39]	2018	1	2	3	1	3	1.5	**11.5**
Fard et al[40]	2018	1	2	3	1	3	1.5	**11.5**
Nguyen et al[29]	2019	1	2	3	0	3	1.5	**10.5**
Gorantla et al[69]	2019	1	2	3	0	3	1.5	**10.5**
Ershad et al[41]	2019	1	2	3	0	3	1.5	**10.5**
Kowalewski et al[25]	2019	1	2	3	2	3	1.5	**12.5**
Funke et al[68]	2019	1	2	3	0	3	1.5	**10.5**
Fawaz et al[42]	2019	1	2	3	0	3	2	**11**
Anh et al[43]	2020	1	2	3	0	3	1.5	**10.5**
Khalid et al[44]	2020	1	2	3	0	3	1.5	**10.5**
Loukas et al[63]	2020	1	2	3	0	3	1.5	**10.5**
Zhang et al[81]	2020	1	2	3	0	3	1.5	**10.5**
Kelly et al[79]	2020	1	3	3	2	3	1.5	**13.5**
Gahan et al[70]	2020	1	2	3	2	3	1.5	**12.5**
Brown et al[45]	2020	1	2	3	1	2	1.5	**10.5**
SIMULATION								
Megali et al[50]	2006	1	2	3	0	3	1.5	**10.5**
Ahmidi et al[51]	2012	1	2	3	2	3	1.5	**12.5**
Zhu et al[83]	2015	1	2	3	3	3	1.5	**13.5**
Poursartip et al[52]	2017	1	2	3	2	3	1.5	**12.5**
Topalli et al[53]	2019	1	2	3	0	3	1.5	**10.5**
Winkler-Schwartz et al[54]	2019	1	2	3	0	3	1.5	**10.5**
Peng et al[55]	2019	1	2	3	1	3	1.5	**11.5**
Bissonnette et al[82]	2019	1	3	3	1	3	1.5	**12.5**
Siyar et al[56]	2020	1	2	3	0	3	1.5	**10.5**

Mirchi et al[57]	2020	1	2	3	0	3	1.5	**10.5**
REAL								
Ahmidi et al[49]	2015	1	2	3	0	3	1.5	**10.5**
Jin et al[71]	2018	1	2	3	2	3	1.5	**12.5**
Hung et al[47]	2018	1	2	3	2	3	3	**14**
Baghdadi et al[72]	2019	1	2	3	2	3	1.5	**12.5**
Kim et al[74]	2019	1	2	3	3	3	1.5	**13.5**
Azari et al[75]	2019	1	2	3	3	3	1.5	**13.5**
Lee et al[73]	2020	1	2	3	2	3	1.5	**12.5**
Chen et al[48]	2020	1	2	3	0	3	1.5	**10.5**